Lecture 7: Data as Specification

CS120: Introduction to AI Safety - October 13, 2026, Zachary Robertson

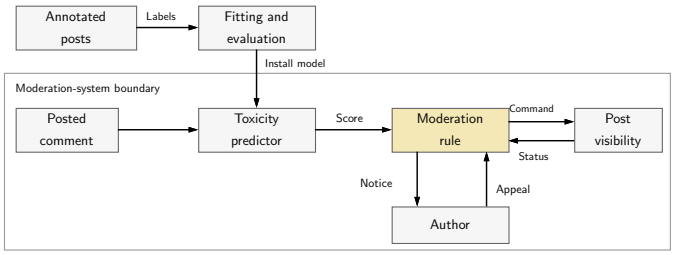
Who specifies the task?

Generalization	Performance on future samples from a specified population.
Robustness	Behavior under specified changes in inputs or conditions.

Who chose the population, labels, groups, and permitted changes?

Today: trace those choices into data production and model selection.

A score does not remove a post



Intended policy: remove targeted abuse while allowing other comments.



Roadmap

Whose judgments become labels?

Document consequential choices

Change data and trace consequences

A recorded rating is not the intended policy

Recorded judgment	How offensive or toxic does an annotator perceive this comment to be?
Intended decision	Does the comment violate the platform's policy against targeted abuse?

Accurate prediction of ratings does not establish accurate application of the policy.

Annotators with Attitudes

Sap et al. study perceived offensiveness and its associations with dialect, prejudice, and vulgarity.

Studied comments include African American English (AAE), racism, and vulgarity.

Higher racist-belief scores	Associated with lower toxicity ratings for anti-Black posts.
The same attitude measure	Associated with higher toxicity ratings for AAE posts.

Sap et al., NAACL 2022

Where can a dialect shortcut originate?

A possible mechanism upstream of model fitting:

Annotator composition and judgments → Dialect-toxicity associations in labels → A predictor that relies on dialect

A dataset can reward the association before a model learns it.

Researchers also make specification choices

Breadth-of-posts study: 173 annotators label 571 tweets.

Measurement	One questionnaire item per attitude dimension, selected using the earlier study.
Assignment	Two White conservatives, two White liberals, and two Black annotators per post.

Which comparisons and measurement claims does this design support?

Sap et al., section 3.2 and Appendix D

What did the study establish?

Evidence reported	Associations among measured attitudes, identities, and annotations in the study.
Further model claim	Training on those labels produces a predictor that relies on dialect.
Further system claim	That predictor causes permitted posts to be removed with consequential delays.



Roadmap

Whose judgments become labels?

Document consequential choices

Change data and trace consequences

Datasheets make choices traceable

Gebru et al. propose documenting how a dataset was made and how it should be used.

Purpose and composition	Why this dataset? What does it contain?
Collection process	How were instances and judgments obtained?
Processing and labeling	Which transformations produced the usable data?
Uses	Which uses are recommended or discouraged?

Gebru et al., Datasheets for Datasets

Documented, inferred, or unknown?

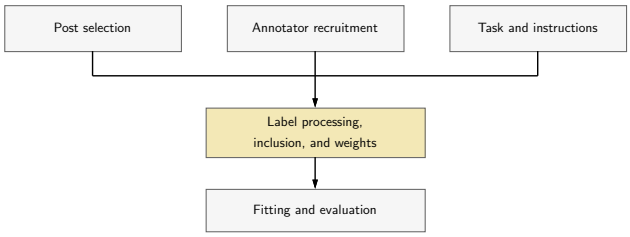
Documented	An attributed claim about what was done or observed.
Inferred	A conclusion requiring reasoning and explicit assumptions.
Unknown	A question the available sources do not answer.

An unanswered question does not establish that a step was omitted.

Follow provenance into a reuse decision

Source	Ratings elicited to measure perceived offensiveness.
Assumption	Those ratings can stand for violations of the intended policy.
Check	Compare ratings with independently justified policy judgments.
Decision	Substantial mismatch may require new labels or a narrower use.

Expand the development boundary



Documentation guides inquiry; it does not certify safety

Which fact or empirical result would change the decision to use this data?

Can support	Tracing choices, exposing limitations, and targeting further investigation.
Does not establish alone	Construct validity, acceptable predictor behavior, or timely correction of removals.



Roadmap

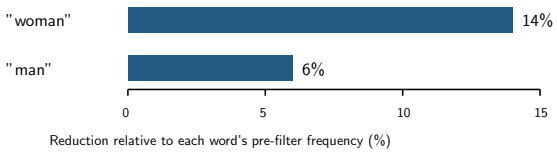
Whose judgments become labels?

Document consequential choices

Change data and trace consequences

Filtering can change more than its target

DALL-E 2 pre-training: filter sexual and violent imagery to reduce problematic outputs.



OpenAI, DALL-E 2 Pre-Training Mitigations, 2022

Reweighting is a second intervention

Filtering	Changes which examples are retained.
Reweighting	Changes the influence of retained examples during fitting.

A control aimed at one problem can motivate a further mitigation.

OpenAI, DALL-E 2 Pre-Training Mitigations, 2022

Data choices enter the training objective

$$\hat{\theta}_c \in \arg\min_{\theta \in \Theta} \frac{\sum_{i \in I_c} w_i^{(c)} \ell(\theta; x_i, y_i^{(c)})}{\sum_{i \in I_c} w_i^{(c)}}$$

I_c	Which instances are retained
$y_i^{(c)}$	Which label is used
$w_i^{(c)}$	How much each instance contributes

Weights are nonnegative and their total is positive. Here, i can index annotations.

Choose the policy, then fit the model

Choose development policy c → Fit parameters $\hat{\theta}_c$ → Evaluate application behavior

Intended change	Which measured problem should the intervention reduce?
Intended invariance	Which relevant behavior should remain acceptable?

First fix the baseline pipeline

Hypothetical harm scenario

Fixed policy c_0	Repeated training runs return context-sensitive and dialect-shortcut predictors.
Validation equivalence	Their validation performance is statistically indistinguishable.
Unresolved behavior	Validation omits non-abusive AAE; the predictors differ on permitted posts.

Then change the development policy

Change $c_0 \to c_1$; keep the same validation check.

Treat one attitude item as a reliability measure → Upweight high-scoring annotators → Select a dialect-shortcut predictor

Assume the pilot confounds dialect with toxicity and those annotators rate dialect itself as toxic.

Changing the objective does not repair the validation gap.

From a label association to lost participation

The deployed predictor encounters an AAE post the policy permits.

Predictor flags the permitted post → Moderation rule issues a removal → Appeal restores it too late

Hazard H_T	An allowed post is hidden while timely correction is unavailable.
Harm L_T	The speaker loses timely participation because the post was removed.

Requirements need evidence at different points

SC_T**: prevent or correct erroneous removals before the participation window closes.**

Development check	Evaluate permitted non-abusive AAE before and after reweighting; compare validation-equivalent predictors.
System check	Compare removal, appeal, and restoration times with a defined participation window.

Data specifies learning, not safety by itself

Provenance	Makes data-production choices and uncertainty traceable.
Validation	Tests specified behavior; other differences may remain unresolved.
Selection	Determines which predictor is installed.
System controls	Shape whether prediction errors become participation harm.

Discussion: should the platform reuse the breadth-of-posts data?

Discussion: Data as specification

A moderation team is considering reusing the *breadth-of-posts* data from Sap et al., *Annotators with Attitudes*, to train the toxicity predictor from lecture. The platform intends to remove targeted abuse while allowing other comments. Use Sap et al. §3.2 and Appendix D.1–D.2, and Gebru et al., *Datasheets for Datasets*, §3.2 (Composition), §3.3 (Collection Process), and §3.5 (Uses). If the sources do not answer a question, mark it **unknown** rather than inferring an answer. Do not use the lecture's one-item attitude/reweighting example.

1. Proposed reuse

What did Sap et al. collect the ratings to study, and what would the platform ask those ratings to stand for? State one assumption needed for this reuse.

2. Composition and sampling

For the posts, annotators, and ratings, what are the dataset's instance types and how were they sampled? What larger population, if any, can each reasonably represent?

3. Consequential provenance

Choose one documented collection choice or one **unknown** from Question 2. Give a plausible route by which it could contribute to a permitted post being removed. What additional assumption does your route require?

4. Reuse decision

Using *Datasheets* §3.5, identify one additional fact or empirical check that would most affect reuse. What result would make you more or less willing to reuse the data, and what safety claim would still remain unsupported? Conclude: **reuse / reuse with conditions / do not reuse yet**.